FOOT & ANKLE · NEET-SS / FET / INI-SS

# Hallux valgus and forefoot

## Part A: Hallux valgus

### 1. Definition and aetiology

- **Lateral deviation of the hallux at the first MTPJ with medial deviation of the first metatarsal (metatarsus primus varus)**, with a medial prominence ("bunion").
- **Women (about 10:1 in surgical series)**, **familial (about 60-90% of juvenile cases; maternal transmission)**, **constricting footwear (narrow toe box, high heels)**, **ligamentous laxity**, **pes planus**, **first ray hypermobility (debated)**, **long first metatarsal**, **rounded metatarsal head**, **increased DMAA**, **inflammatory arthritis (RA)**, **neuromuscular (cerebral palsy, stroke)**, **amputation of the second toe**.

### 2. Pathoanatomy (sequence)

1. **Medial capsule and medial sesamoid ligament attenuation**; **the hallux drifts laterally (valgus)** pulled by **adductor hallucis** and the lateral pull of the **EHL and FHL** (**bowstring effect: become deforming forces**).
2. **The metatarsal head drifts medially off the sesamoids** (the **sesamoids stay in place: anchored by the adductor hallucis and the transverse metatarsal ligament**): **apparent lateral sesamoid displacement** relative to the metatarsal head; **crista erosion**.
3. **Pronation (eversion) of the hallux** (abductor hallucis migrates plantarly: loses abduction power; becomes a flexor/pronator).
4. **Lateral capsule and adductor contracture**; **transfer metatarsalgia** under the second metatarsal head (first ray loses load-bearing).
5. **Second toe deformities** (crossover toe, hammer toe), **second MTPJ instability/plantar plate tear**.

## 3. Evaluation

- **Weight-bearing AP and lateral radiographs** (and sesamoid axial view).
- **Normal values and severity:**

| Angle | Normal | Mild | Moderate | Severe |
|---|---|---|---|---|
| **Hallux valgus angle (HVA: first MT axis vs proximal phalanx axis)** | **Under 15 degrees** | **15-20** (up to 20-25 in some schemes) | **20-40** | **Over 40** |
| **Intermetatarsal angle (IMA: 1st-2nd MT)** | **Under 9 degrees** | **9-11** (under 13) | **11-16** (13-20) | **Over 16** (over 20) |
| **Distal metatarsal articular angle (DMAA)** | **Under 10 degrees** (about 6-10) | | | |
| **Hallux valgus interphalangeus (HVI)** | **Under 10 degrees** | | | |

- **Congruent vs incongruent joint: congruent (articular surfaces parallel; increased DMAA; juvenile)** vs **incongruent (subluxed: lateral subluxation of the phalanx on the head)**. **Congruent deformity: correction must not be done by soft-tissue release alone (would create an incongruent joint): use DMAA-correcting osteotomy (biplanar chevron) or extra-articular (double/triple) osteotomies**.
- **Sesamoid position (Hardy and Clapham: grades 1-7)**, **first MTPJ arthritis** (**hallux rigidus: fusion instead**), **first TMT hypermobility** (**dorsal translation over about 8-10 mm: Lapidus**), **metatarsus adductus** (**increased difficulty; IMA underestimated**), **pronation of the first metatarsal (round lateral head sign)**.

## 4. Management

### Non-operative

- **Wide toe-box footwear, bunion pads, orthoses, toe spacers**; **do not correct the deformity**; **surgery is for pain, not cosmesis** ("**never operate for cosmesis**").

## Operative principles: choose by severity, joint congruency, arthritis and TMT stability

| Deformity | Procedure | Notes |
|---|---|---|
| **Mild (HVA under 30, IMA under 13), congruent or incongruent** | **Distal metatarsal osteotomy: Chevron (Austin)** (**V-shaped osteotomy at the metatarsal head, apex distal; lateral shift up to about 30-50% of the head width**) ± **distal soft-tissue procedure (lateral release)** | **AVN of the metatarsal head** (**lateral release + chevron: avoid excessive dorsolateral dissection: nutrient vessels enter dorsolaterally (first dorsal metatarsal artery) and plantar-laterally**); **biplanar chevron for increased DMAA** |
| **Moderate (HVA 30-40, IMA 13-20)** | **Scarf osteotomy** (**Z-shaped diaphyseal osteotomy**: lateral translation; **troughing (fragments telescope into each other: elevation of the head, loss of height)**), **proximal metatarsal osteotomy (proximal chevron, crescentic, opening/closing wedge, Ludloff, Mau)** + **distal soft-tissue release** (modified McBride) | **Proximal osteotomies: more powerful correction (each degree of correction at the base produces more displacement distally)**; **dorsal malunion (elevatus: transfer metatarsalgia)** |
| **Severe (HVA over 40, IMA over 20)** | **Proximal osteotomy + distal soft-tissue procedure**, **Lapidus (first TMT arthrodesis)** | |

| Deformity | Procedure | Notes |
|---|---|---|
| **First TMT hypermobility / generalised laxity / recurrent HV** | **Modified Lapidus (first TMT arthrodesis)** | **Non-union (about 5-10%)**, **shortening (transfer metatarsalgia)**, dorsiflexion malunion |
| **Arthritis of the first MTPJ, severe deformity, neuromuscular, RA, salvage of failed surgery** | **First MTPJ arthrodesis** | **Position: about 10-15 degrees valgus, about 10-15 degrees dorsiflexion relative to the floor (about 20-25 degrees relative to the first metatarsal shaft), neutral rotation**; **IMA corrects spontaneously** after fusion |
| **Hallux valgus interphalangeus** | **Akin osteotomy** (**medial closing-wedge osteotomy of the proximal phalanx**) | Adjunct to other osteotomies |
| **Elderly, low-demand** | **Keller resection arthroplasty** (**excision of the base of the proximal phalanx**) | **Cock-up deformity, transfer metatarsalgia, weakness of push-off**; largely abandoned |
| **Juvenile/adolescent** | **Delay until skeletal maturity** (**high recurrence**); **guided growth (lateral hemiepiphysiodesis of the first metatarsal)** | **Increased DMAA common** |

- **Simple bunionectomy (Silver) alone**: **high recurrence; not used alone**.
- **McBride procedure** (original): **excision of the lateral sesamoid** (**hallux varus risk**), **adductor tenotomy**, **medial capsulorrhaphy**: **modified McBride = lateral release without sesamoid excision**.

- **Minimally invasive (MIS) third/fourth generation (MICA/percutaneous chevron-Akin)**: increasingly used; similar outcomes; learning curve.

### 5. Complications

- **Recurrence (commonest; about 10-25%)**: **undercorrection of the IMA**, **unrecognised increased DMAA**, **unaddressed pronation**, **first TMT hypermobility**, **juvenile**, **poor sesamoid reduction**.
- **Hallux varus** (**overcorrection: excessive lateral release, lateral sesamoid excision, excessive medial eminence resection (staking the sagittal sulcus), overtightened medial capsule, IMA overcorrection (negative)**): **flexible: EHB tenodesis/EHL transfer; rigid/arthritic: MTPJ fusion**.
- **AVN of the metatarsal head** (**chevron + extensive lateral release**).
- **Transfer metatarsalgia** (**shortening or dorsal malunion of the first metatarsal**).
- **Stiffness of the first MTPJ**, **non-union/malunion**, **infection**, **dorsomedial cutaneous nerve injury (neuroma)** (**medial dorsal cutaneous branch of the superficial peroneal nerve**), **CRPS**.

## Part B: Hallux rigidus

### 6. Hallux rigidus (first MTPJ OA)

- **Commonest arthritis of the foot**; **dorsal osteophyte (dorsal bunion)** limiting **dorsiflexion (pain at push-off)**; **positive grind test (late)**; **adolescent form (OCD of the metatarsal head)**.
- **Coughlin and Shurnas classification (grades 0-4): 0 stiffness, normal radiograph**; **1 mild, dorsal osteophyte, preserved joint space**; **2 moderate, joint space narrowing, pain at extremes**; **3 severe, under 10-20% residual motion, pain at extremes, more narrowing**; **4: as 3 with pain in mid-range (positive grind test)**.
- **Treatment:**

- **Non-operative: stiff-soled shoe, rocker sole, Morton's extension (carbon plate)**, NSAIDs, steroid injection.
- **Cheilectomy** (**excision of the dorsal osteophyte and dorsal 25-30% of the metatarsal head**): **grades 1-2 (and selected 3)**; **pain at the extremes, not mid-range**; **± Moberg osteotomy (dorsal closing-wedge osteotomy of the proximal phalanx: shifts the arc into dorsiflexion)**.
- **First MTPJ arthrodesis**: **gold standard for grades 3-4** (**predictable pain relief**; **position: 10-15 degrees dorsiflexion relative to the floor, 10-15 degrees valgus, neutral rotation**).
- **Interpositional arthroplasty/Keller** (elderly), **hemiarthroplasty (synthetic cartilage implant, Cartiva: mixed results)**; **silicone/total joint arthroplasty: high failure (silicone synovitis, subsidence)**.

# Part C: Lesser toe deformities and forefoot pain

## 7. Lesser toe deformities

| Deformity | MTPJ | PIPJ | DIPJ | Notes |
|---|---|---|---|---|
| **Mallet toe** | Neutral | Neutral | **Flexed** | **FDL contracture** |
| **Hammer toe** | **Neutral or slightly extended** | **Flexed** | **Neutral/extended** | **Commonest; second toe**; FDL/FDB contracture |
| **Claw toe** | **Hyperextended** | **Flexed** | **Flexed** | **Usually multiple toes, bilateral: neuromuscular (CMT, diabetes neuropathy), RA, cavus foot: intrinsic weakness** (lumbrical/interossei) |

| Deformity | MTPJ | PIPJ | DIPJ | Notes |
|---|---|---|---|---|
| **Curly toe** | | Flexed + varus (rotated) | | **Congenital** (FDL tightness) |

- **Flexible deformities: flexor-to-extensor transfer (Girdlestone-Taylor: FDL split to the extensor hood/dorsum of the proximal phalanx)**, **FDL tenotomy (mallet)**.
- **Fixed deformities: PIPJ resection arthroplasty (DuVries) or PIPJ arthrodesis**; **MTPJ: extensor tendon lengthening (EDL Z-lengthening), EDB tenotomy, dorsal capsulotomy**; **Weil osteotomy** (**oblique shortening osteotomy of the metatarsal neck**: for **MTPJ dislocation/subluxation, long metatarsal**: **floating toe** = commonest complication).
- **Crossover second toe / second MTPJ instability (plantar plate tear): dorsomedial drift of the second toe**, **pain under the second MTPJ**, **positive drawer (Lachman) test of the MTPJ**; **Coughlin plantar plate grading**; **plantar plate repair + Weil osteotomy**, flexor-to-extensor transfer.
- **Freiberg disease** (second metatarsal head osteochondrosis: adolescent girls; dorsal closing-wedge osteotomy).

## 8. Metatarsalgia

- **Pain under the metatarsal heads**: **primary** (**long second metatarsal**, plantarflexed metatarsal, **cavus foot**, prominent condyles), **secondary** (**inflammatory arthritis, Freiberg, MTPJ synovitis, gastrocnemius tightness**), **iatrogenic (transfer after hallux valgus surgery)**.
- **Intractable plantar keratosis (IPK)**: **discrete** (beneath a prominent fibular condyle) vs **diffuse** (beneath a long metatarsal).
- **Gastrocnemius contracture (positive Silfverskiöld test: ankle dorsiflexion limited with the knee extended, improves with the knee flexed)**: **gastrocnemius recession (Strayer)**.
- **Treatment: metatarsal pad (placed proximal to the metatarsal heads)**, rocker-bottom sole, orthoses; **Weil osteotomy**, **condylectomy**.

## 9. Morton neuroma (interdigital neuroma)

- **Perineural fibrosis of the common digital nerve** (**not a true neuroma**), **distal to the deep transverse metatarsal ligament**; **third web space (between 3rd and 4th metatarsal heads) commonest**, then second; **women (about 5-10:1)**, **narrow/high-heeled shoes**.
- **Burning plantar forefoot pain radiating into the toes**, **numbness**, **"walking on a pebble"**; **Mulder click (squeezing the forefoot mediolaterally while pressing the web space)**, **web space tenderness**; **splaying of the toes (Sullivan sign)**.
- **Imaging: ultrasound or MRI** (size **over about 5 mm**); **differentials: MTPJ synovitis/plantar plate tear (second web), stress fracture, intermetatarsal bursitis, tarsal tunnel syndrome, lumbar radiculopathy**.
- **Treatment: wide shoes, metatarsal pad, steroid injection (up to about 50-80% relief)**, **alcohol sclerosing injections**, **radiofrequency ablation**; **surgery: neurectomy (dorsal approach preferred: avoids plantar scar; divide the deep transverse metatarsal ligament; resect the nerve proximal to the ligament, about 3 cm)** (**about 80-90% success**); **recurrent (stump) neuroma (about 5-10%)**: **plantar approach** for revision; **decompression (ligament release) alone** as an alternative.

## 10. Sesamoid disorders

- **Sesamoiditis, stress fracture, AVN, osteochondritis, OA, bipartite sesamoid (medial/tibial sesamoid commonest: about 10-30%; bilateral in about 85%: compare with the other foot)**, **turf toe**.
- **Turf toe: hyperdorsiflexion injury of the first MTPJ (plantar plate/capsuloligamentous complex)**: **American football on artificial turf**; **grade I** sprain, **II** partial tear, **III** complete tear (**proximal migration of sesamoids**, **sesamoid diastasis**, **vertical instability: Lachman positive**); **grade III: surgical repair** of the plantar plate.
- **Treatment of sesamoid pain:** offloading (dancer's pad, orthoses), boot/cast; **sesamoidectomy** (refractory: **excision of both sesamoids**

**causes cock-up deformity: avoid**; **medial sesamoid excision: hallux valgus**; **lateral sesamoid excision: hallux varus**).

## 11. Bunionette (tailor's bunion)

- **Lateral prominence of the fifth metatarsal head**; **Coughlin types: I enlarged/prominent head**, **II lateral bowing of the fifth metatarsal**, **III increased 4-5 intermetatarsal angle (over 8-10 degrees: commonest symptomatic)**.
- **Treatment: wide shoes**; **type I: lateral condylectomy** or **distal chevron osteotomy**; **type II: distal chevron or diaphyseal (oblique/scarf) osteotomy**; **type III: diaphyseal oblique or proximal osteotomy** (**proximal fifth metatarsal osteotomy: poor blood supply: non-union**).

Exam pearl: **HVA under 15 and IMA under 9 normal; DMAA under 10. Sesamoids stay; the metatarsal head drifts medially. Congruent joint (increased DMAA, juvenile) needs DMAA-correcting osteotomy, not soft-tissue release. Mild: chevron; moderate: scarf/proximal osteotomy; severe/hypermobile: Lapidus; arthritic: fusion (10-15 valgus, 10-15 dorsiflexion to floor). Commonest complication: recurrence; hallux varus = overcorrection (lateral sesamoid excision).**

Exam pearl: **Hallux rigidus: Coughlin-Shurnas; cheilectomy for pain at the extremes (grade 1-2, ± Moberg); mid-range pain/grind (3-4) = fusion.**

Exam pearl: **Claw toe (MTPJ hyperextension): neuromuscular/cavus; flexible: Girdlestone-Taylor (FDL to extensor). Crossover second toe = plantar plate tear (drawer test): repair + Weil (floating toe). Morton neuroma: 3rd web, Mulder click, distal to DTML; neurectomy via dorsal approach. Bunionette type III (increased 4-5 IMA) = diaphyseal osteotomy.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.